Structural Self-Energy Expansion (SSEE)

A minimal-parameter dark energy framework from φ and π

Mike Edison Almeida Vallejo ORCID: 0009-0008-2195-7836
mike.almeida1721@gmail.com github.com/mikealmeida1721/SSEE-Cosmologia

Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

The dimensionless core of the cosmological background — w_0, w_a, Ω_m, n_s — is derived *algebraically* from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π with **zero fitted dimensionless parameters at the background level**; the one dimensionful input is the measured local rate (SH0ES), from which the global H_0 follows. The kineticity $\alpha_K = 15.591336$ of the dark-energy condensate (Paper 7) is derived, not fitted, and is unobservable on linear scales. The dark-sector ansätze that previously appeared here (m_φ, ξ) were retracted in 2026-08 together with the second matter sector, and their register entries (OP-9 through OP-12) close by dissolution; the MIRA matter mechanism (OP-8) had already dissolved under the ω_m-direct reframe. The remaining open entries are listed in `OPEN_PROBLEMS.md`.

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agreement
w_0	$-T_r/M_v = -0.839950$	DESI DR2+CMB+Pantheon+: -0.838 ± 0.055	-0.04σ
w_a	$-P_{sc}/I_g = -0.669975$	DESI DR2+CMB+Pantheon+: -0.617 ± 0.208	-0.26σ; joint 2D: 0.24σ
$3(\varphi+\pi)^2$ (target)	pure number $= 67.96214$	Planck 2018: 67.36 ± 0.54	1.12σ
$\Omega_{m,\mathrm{CMB}}$	$\omega_m/h^2 = 0.308881$ (ω_m-direct)	Planck 2018: 0.3153 ± 0.0073	0.88σ
r_d	CAMB (ω_m-direct): 147.17 Mpc	Planck 2018: 147.09 ± 0.26 Mpc	0.32σ
H_0^{glob}	$H_0^{\mathrm{SH0ES}}(1-f_{\mathrm{scr}}) = 67.962142$ km/s/Mpc	$3(\varphi+\pi)^2 = 67.96214$	residual $+4.2\times10^{-6}$ ($\sigma = 0.968$)
n_s	$1-\varphi^{-7} = 0.965558$	Planck 2018: 0.9649 ± 0.0042	0.16σ
r	$\varphi^{-10} = 0.008131$	BK18: $r < 0.036$	Prediction
c_s^2	$(M_v - T_r)/(5M_v + 3T_r) = 0.021284$	dark-energy clustering (Euclid)	Prediction
S_8	0.8273 (predicted; *no* free cosmological parameter)	KiDS-Legacy: 0.8265 ± 0.0176	0.04σ

The result in one line: it ties everywhere, and DESI separates

With the background of both models fixed in advance, nothing fitted outside the CMB and only $\{A_s, \tau\}$ in it, SSEE ties ΛCDM probe by probe: $|\Delta\chi^2| \leq 3.0$ in 10 of 11 comparisons (supernova compilations, ACT and SPT CMB lensing, DES Y3, Planck, KiDS-Legacy), even where ΛCDM fits up to 6 parameters against SSEE's $\{A_s, \tau\}$; BOSS full-shape gives $\Delta\chi^2 = +2.84$ against ΛCDM with its cosmology free. DESI DR2 BAO is the only probe that separates the models: $\chi^2 = 11.41$ for SSEE against 31.98 for ΛCDM at Planck ($\Delta\chi^2 = -20.6$, 13 points). Supernova rows share objects and CMB-lensing rows share sky, so they are not summed.

Ten-paper series (all reproducible)

I. Derived Algebraic Background

- **Paper 1 (Framework)**: Algebraic derivation of geometric constants; α-attractor embedding ($\alpha = \varphi^4/3$). [35 pp]
- **Paper 4 (Two-Axiom Cosmology)**: CMB observables (ℓ_1, ℓ_2, ℓ_3, θ_*, A_s, n_s, τ, r) bounded algebraically. [19 pp]

II. Tested Against Data

- **Paper 2 (MCMC)**: 100 walkers × 25 000 steps vs. DESI DR2 + Planck (the cluster term was removed: cluster masses are tested separately, in general relativity, against the 46 systems of Zhang et al. 2026); $\Delta\mathrm{BIC} = -10.90$ with nothing free ($k_{\mathrm{SSEE}} = 0$ vs. $k_{\Lambda\mathrm{CDM}} = 3$; -5.81 in the conservative version that samples H_0 and $\Omega_b h^2$), total matter density in the background geometry. [28 pp] Convention throughout this summary: $\Delta \equiv$ SSEE − ΛCDM; negative Δ favours SSEE.

- **Paper 3 (CMB Confrontation)**: CAMB ω_m-direct ($\Omega_m = 0.308881$); Cobaya full `plik` MCMC; $\Delta\mathrm{BIC} = -33.83$ (canonical; -26.03 in the `plik_lite` point estimate with $\{A_s, \tau\}$ fitted and $N = 669$ measured from the likelihood), SSEE favoured under $k = 2$ vs $k = 6$ model comparison. [24 pp]

- **Paper 6 (Growth sector, against raw $\xi_\pm$)**: retracts an earlier two-sector φ-DM proposal and its particle — the subtraction defining the second sector mixed a measured density with an equation-of-state number, and the S_8 tension motivating it exists only in the compressed statistic, whose reduction assumes ΛCDM. Against the **KiDS-Legacy** release (357 raw points; Wright et al. 2025, arXiv:2503.19441 — *published sixteen months before this analysis: no temporal priority is claimed*) the shear asks for $\log(10^{10}A_s) = 3.0544 \pm 0.0383$ against the 3.04483 that the CMB fixes under the *same* algebraic background: -0.25σ (Planck background, held equally rigid: 0.16σ). Fixing A_s to that CMB value therefore costs almost nothing — $\chi^2 = 415.59$ with **eight free parameters, all nuisance, none cosmological**, against 414.99 with A_s free ($\Delta\mathrm{BIC} = +5.28$) and 415.74 for the ΛCDM-background control ($+6.02$). $S_8 = 0.8273$ is then predicted, not fitted, against 0.8265 ± 0.0176 measured: 0.04σ. The three χ^2 span 0.74, so the shear does *not* discriminate between the models; what separates them is the parameter count. (Earlier, against KiDS-1000: $S_8 = 0.7559 \pm 0.0189$ with A_s free, matching the ΛCDM control 0.7571 ± 0.0194 run through the same pipeline with four more free parameters.) [17 pp]

III. Phenomenological Extensions & Open Directions

- **Paper 5 (Israel–Stewart Perturbations)**: Causal viscosity; the vanishing high-frequency sound speed is *not* a stability result — the dark-energy sector is stabilised by the sound speed of the field of Paper 7; $\gamma_{\mathrm{IS}} = 0.5504 \approx \gamma_{\Lambda\mathrm{CDM}}$. [29 pp]

- **Paper 7 (Canonical EFT Proposal)**: the action is a minimally coupled ghost condensate, $K(X) = c_1 X + c_2 X^2$, with no potential and no dark coupling; $\alpha_T = \alpha_M = \alpha_B = 0$, $\alpha_K = 15.591336$ (not observationally accessible) and sound speed $c_s^2 = 0.021284$. The same action fixes w_0 and c_s^2 today but not w_a: the field rolls towards $w = -1$ and cannot cross it (OP-24). [18 pp]

IV. Strong-Field Regime, Hubble Tension & UV Completion

- **Paper 8 (GR Lensing in the Canonical Limit)**: its canonical lensing prediction rests on $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ and $\alpha_B = \alpha_M = 0$ (Paper 7), and is unconditional. In the canonical limit the residual fifth force is $\sim 6 \times 10^{-13}$ at 10 kpc and the observable lensing is that of GR with the model's cold matter; a disformal contrast (deflection $\sqrt{\mathrm{AURA}} = 1.9995$) is kept only for comparison and makes no prediction. Baryons do not couple to the scalar, so solar-system tests are satisfied. Current SLACS/BELLS data are consistent with the canonical limit (OPEN_PROBLEMS.md OP-13). No screening radius is quoted: an earlier Vainshtein/k-mouflage radius formula was dimensionally inconsistent and has been withdrawn (OP-4, closed 2026-10-03); the canonical-limit prediction never used it and rests on ω_c and on $\alpha_B = \alpha_M = 0$. [17 pp]

- **Paper 9 (Hubble Tension)**: Algebraic local screening fraction $f_{\mathrm{scr}}^{\mathrm{IR}} = (\pi - \varphi)/(\varphi + \pi)^2 = 0.067253$ applied to the measured SH0ES rate 73.04 ± 1.04 returns, with the IR term alone, $H_0^{\mathrm{glob,IR}} = 68.13$ km/s/Mpc, 0.17σ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$, the comparison target (SH0ES is the input; no fitted parameters; the multiplicative screening is

a postulate). With the UV term of Paper 10 the same cascade gives the single global rate $H_0^{\rm glob} = 67.96214$. [22 pp]

- **Paper 10 (UV Completion)**: Higher-derivative operator $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ whose cutoff $M = 9.68$ meV is conjectured to be set by the inflationary α-attractor (Conjecture C.1); the Hubble cascade is consistent with it but does not measure it. [16 pp]

Falsifiability (explicit)

- $S_8 = 0.8273$ — with the background fixed by the algebra *and* A_s fixed to the value the CMB gives under that same background, the growth sector has no free cosmological parameter left: cosmic shear placing S_8 more than 3σ away falsifies it outright, with nowhere to absorb the discrepancy (KiDS, DES, Euclid).
- $\log(10^{10}A_s)$ — the amplitude the shear asks for must keep agreeing with the one the CMB fixes. The earlier, superseded KiDS-1000 release asked for 2.864 ± 0.050 (3.6σ away) and KiDS-Legacy for 3.0544 ± 0.0383 (-0.25σ); the two releases differ from each other by 3.0σ. A future release returning to the lower value reopens the sector.
- $\alpha_T = \alpha_M = \alpha_B = 0$ (Paper 7) — gravity is unmodified, $G_{\rm eff} = G_N$ and no gravitational slip; a detected slip or $\mu \neq 1$ in Euclid/LSST lensing×clustering would falsify the sector.
- $r = \varphi^{-10} = 0.008131$ — LiteBIRD will confirm or refute the tensor-to-scalar ratio.
- A $> 3\sigma$ deviation of (w_0, w_a) from $(-0.840, -0.670)$ in DESI DR3 would falsify the corresponding background sector of the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and LaTeX sources are public at github.com/mikealmeida1721/SSEE-Cosmologia under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck 2018 (public), KiDS-1000 and KiDS-Legacy (public).